

To Profit or not to Profit? Evidence from Chile's For-profit Voucher Schools

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Abstract

This note examines whether for-profit management affects educational outcomes in Chile's voucher system. Using administrative panel data for all voucher-subsidized primary schools from 2010 to 2014, we compare for-profit and nonprofit schools in terms of inputs and student performance. For-profit schools serve more disadvantaged students and rely on lower-cost inputs, such as less-qualified teachers, but they also provide more instructional hours and smaller classes. After controlling for student characteristics, a significant test-score gap remains, but it is largely explained by differences in school inputs. These results suggest that ownership matters mainly through differences in how schools allocate educational inputs.

Keywords: for-profit schools; voucher system; education policy; school choice

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All errors are our own.

1 Introduction

For-profit involvement in education remains contested. Critics argue that profit-maximizing behavior may reduce service quality when firms are entrusted with public resources, particularly in compulsory schooling (Chung, 2012; Cellini & Chaudhary, 2014; Armona et al., 2022). Empirical evidence, however, remains scarce and mixed: some studies find comparable or superior performance in for-profit schools, while others document weaker outcomes or behavior consistent with cream-skimming or input substitution (Singleton, 2017; Sahlgren, 2011; Sánchez, 2025).

Chile offers a useful setting in which to study these issues. Since 1981, its universal voucher system has funded public and private schools on a per-student basis, with parents free to choose schools regardless of residence. Roughly two-thirds of private subsidized schools operated under for-profit status during our sample period, yet they faced the same curricular and accountability requirements as nonprofit schools.¹ This institutional setting allows a direct comparison of nonprofit and for-profit schools under a common funding and regulatory regime.

Using administrative panel data for the universe of voucher-subsidized primary schools from 2010 to 2014, we estimate panel regressions relating school inputs and test scores to an indicator for for-profit management. Our preferred specifications include municipality-by-year fixed effects and controls for student composition and school characteristics. We find that for-profit schools serve more disadvantaged students and combine lower-cost teacher inputs with lower pupil-teacher ratios, more instructional hours, and smaller classes. The raw nonprofit advantage in test scores narrows substantially once student composition is taken into account and is largely explained by differences in school inputs.

This note contributes to the small but growing literature on for-profit schools. Our results complement evidence from Chile, Sweden, Florida, and Ohio showing that performance differences across ownership types depend strongly on institutional context and comparison group (Elacqua, 2011; Sahlgren, 2011; Singleton, 2017; Lavertu & Tran, 2025; Sánchez, 2025). Our main contribution is to show that, in Chile’s voucher system, the nonprofit test-score advantage over for-profit schools is closely linked to differences in student composition and input choices.

2 Data and Empirical Specification

Since 1981, Chile’s school system has operated under a nationwide voucher program. Families are free to choose schools, subject to travel costs and any top-up fees. Public and subsidized private schools receive per-student government funding, which includes both a base voucher and a targeted

¹In 2016, Chile’s Ministry of Education banned the operation of profit-seeking institutions in primary education. Our data predate that reform.

voucher for disadvantaged students (Sánchez, 2026).²

We use administrative panel data on all Chilean primary (grades 1–8) schools receiving voucher subsidies from 2010 to 2014.³ We link school registries, enrollment summaries, teacher censuses, SIMCE 4th-grade standardized test scores, family background surveys, and tax filings that identify schools’ legal profit status. Important is that our for-profit measure comes from legal tax registration with the Chilean Internal Revenue Service rather than from self-reports or indirect proxies.

The final sample includes over 3,500 schools per year, about 93% of national primary enrollment. This covers all public and private voucher-subsidized schools. Key outcomes are the log of pupil-teacher ratio, log instructional hours per student, log average class size, and standardized verbal and mathematics test scores—normalized to have mean zero and unit variance in every year.

Following Singleton (2017), we estimate reduced-form models of the form:

$$y_{jkt} = \beta_0 + \beta_1 \text{ForProfit}_{jk} + X_{jkt} \beta_2 + \nu_k \times \delta_t + \varepsilon_{jkt},$$

where y_{jkt} is school j ’s outcome in municipality k and year t , ForProfit_{jk} is a binary indicator of for-profit management, X_{jkt} is a vector of controls, and $\nu_k \times \delta_t$ denotes municipality-by-year fixed effects. Standard errors are clustered at the municipality level. Identification comes from within-municipality-year comparisons of schools with different management types.

Input regressions control only for student composition: the share of disadvantaged students and average parental education. Test-score regressions additionally include monthly copayment and school inputs—pupil-teacher ratio, instruction hours, class size, multigrade classes, teacher experience, and the share of teachers with an out-of-field diploma. Because these inputs are choice variables, we use them only for descriptive purposes.

3 Results

Table B.1 summarizes raw differences across school types. Columns (1)–(3) show that private schools are larger, less rural, serve more advantaged students, and post higher test scores than public schools. Columns (4)–(6), our main comparison, show that for-profit private schools are smaller, more rural, and lower-tuition than nonprofit private schools; they also use smaller classes and more instructional hours, employ less-qualified teachers, serve more disadvantaged students, and post lower raw test scores.

Regression results in Table B.2 show that for-profit schools differ systematically in inputs. Column (1) reports raw differences, and column (2) adds municipality×year fixed effects, controlling

²The targeted voucher is optional for subsidized providers: participating schools receive an additional per-pupil payment for each eligible student but may not charge those students top-up fees (Neilson, 2025; Sánchez, 2026).

³We ignore private-non-voucher schools because they are not eligible to receive governmental funding.

for time-varying local market unobservables. The patterns are robust: for-profit schools have lower pupil-teacher ratios, more instruction hours per student, and smaller classes, although the magnitudes attenuate with municipality-year fixed effects. When we add student demographic controls in column (3), the estimates remain similar: the for-profit coefficient is -0.072 for the pupil-teacher ratio, 0.070 for instruction hours per student, and -0.163 for class size (all p -values < 0.01).

Table B.3 presents test-score results. Columns (1) and (2) include no controls and municipality-by-year fixed effects only, respectively. Column (3) adds student characteristics—our preferred specification, as these controls are exogenous to school decisions. Column (4) adds monthly tuition. Column (5) further adds school production inputs—pupil-teacher ratio, instructional hours per student, class size, share of multigrade classes, average teacher experience, and the share of teachers with an out-of-field teaching diploma.

Columns (1) and (2) in Table B.3 show that the raw nonprofit advantage in test scores is not primarily driven by time-varying local market unobservables. Adding municipality $\times$ year fixed effects (column (2)) leaves the estimated gaps largely unchanged (verbal: -0.386 in column (1) versus -0.358 in column (2); math: -0.355 versus -0.336), suggesting that differences in where schools operate account for only a small share of the gap. Once student characteristics are controlled for (column (3)), the gap narrows substantially but remains statistically significant (verbal: -0.170, math: -0.148), indicating that student composition explains part, but not all, of the raw difference. Column (4) adds monthly tuition; the coefficients are virtually unchanged, suggesting that price differences do not account for the remaining gap. Column (5) replaces tuition with school production inputs in the set of controls. The coefficient falls to -0.083 ($p < 0.01$) for verbal and -0.036 ($p = 0.17$) for math, indicating that differences in school inputs largely account for the performance gap.

Finally, we emphasize that these results are descriptive rather than causal, since within-municipality comparisons may still reflect unobserved sorting or other school-level factors correlated with ownership.

4 Conclusion

Using administrative panel data for all voucher-subsidized primary schools in Chile from 2010 to 2014, we show that for-profit schools serve more disadvantaged students and operate with a distinct input mix, combining lower-cost teacher inputs with smaller classes and more instructional hours.

Although raw test scores are lower in for-profit schools, these gaps shrink substantially once student composition is accounted for and are largely explained by differences in school input choices. Our evidence adds to the literature by suggesting that the nonprofit advantage documented in related studies may largely reflect differences in educational input choices (Lavertu & Tran, 2025; Sánchez, 2025).

Competing interests

The authors declare that they have no known competing interests that could have influenced the work reported in this paper.

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A Data

We combine several administrative datasets on Chilean students and schools for the years 2010–2014. Specifically, we use:

1. *Registry of schools, 2010–2014.*

These data provide information on schools' municipalities, administrative type, urban status, address, tuition, religious orientation, and the type and level of education offered.

2. *Registry of schools' summary of enrollment, 2010–2014.*

These data provide information on schools' municipalities, administrative type, urban status, male and female enrollment by education type and level, total enrollment by education type and level, total enrollment, the number of single-grade classes by education type and level, the total number of single-grade classes, the number of multigrade classes by education type and level, and the total number of multigrade classes.

3. *Teachers census, 2010–2014.*

These data provide information on teachers' gender, date of birth, education degree, subject specialization, educational institution attended, graduation year, and the duration of their degree programs. They also provide information on the characteristics of all schools in which each teacher is employed (municipality, administrative type, rural status), as well as the teachers' primary and secondary roles (e.g. teacher, principal, supervisor), contract type, contracted hours, teaching hours, experience, tenure, and teaching subject and level.

4. *Registry of schools that participate in the targeted voucher program, 2010–2014.*

These data provide information on the characteristics of schools that participate in the targeted voucher program, including schools' municipalities, administrative type, urban status, the number of disadvantaged students eligible for the targeted voucher subsidy, and the number of students who benefit from the targeted voucher.

5. *Registry of students who are eligible to participate in the targeted voucher program, 2010–2014.*

These data provide information on the characteristics of disadvantaged students who are eligible to participate in the targeted voucher program. They include students' gender, date of birth, program participation status, level of education, grade, single- or double-shift schedule, and the administrative type and urban status of the school attended by the student.

6. *National standardized exams (SIMCE) for 4th graders, school-level, 2010–2013.*

These data provide information on schools' average test scores in verbal, mathematics, and social/natural science, as well as municipality, administrative type, the socioeconomic category of the population served by the school, urban status, and the number of students taking the tests.

7. *4th-grade SIMCE questionnaire for parents and tutors, 2010–2013.*

These data consist of responses to a survey completed by parents and tutors during the days when the national standardized tests are administered. The survey is voluntary, although more than 90% of parents respond to it each year. It provides information on students' household size, home amenities, time use, the total number of books available at home, total monthly household income, parents' and tutors' time use, education, indigenous identification, occupation, health insurance, participation in social programs, reasons for choosing the school, beliefs about the student's future educational attainment, satisfaction with the school, knowledge of the school's average performance in standardized tests, total monthly education-related expenses other than tuition, and the school's admission criteria, tuition, and fees.

8. *For-profit status indicators, 2011 and 2013.*

These data allow us to identify schools' profit status. The for-profit indicator is constructed from schools' tax records.

We combine these datasets to form a five-year panel of schools. We exclude private fee-paying schools because they are not eligible to receive government funding and serve a small, relatively affluent minority of the country.⁴ We use the 2011 for-profit indicator to identify schools' profit status in the years 2010–2012, and the 2013 for-profit indicator to identify schools' profit status in the years 2013–2014.

⁴Private fee-paying schools enroll about 7% of students nationwide and charge fees that are, on average, three times the amount of the base voucher subsidy.

B Tables

Table B.1: Summary Statistics by Type of School

	public (1)	private (2)	diff (3)	nonprofit (4)	for-profit (5)	diff (6)
<i>A. School Characteristics</i>						
enrollment	169.59	297.34	127.75***	388.12	261.39	-126.73***
monthly tuition (CLP)	0	12,187	12,187***	13,322	11,738	-1,584***
secular (%)	50.06	42.95	-7.11***	17.21	53.52	36.31***
participates in targeted voucher (%)	98.38	68.19	-30.19***	67.24	68.57	1.33*
rural (%)	62.56	24.63	-37.93***	16.50	27.84	11.34***
<i>B. School Inputs</i>						
pupil-teacher ratio	11.78	16.93	5.15***	17.89	16.56	-1.33***
instruction hours per student	4.08	2.21	-1.88***	2.06	2.27	0.21***
class size	17.57	26.27	8.70***	29.88	24.84	-5.04***
% multigrade classes	30.70	13.43	-17.27***	6.90	16.02	9.13***
<i>C. Teacher Characteristics</i>						
institution attended: college (%)	86.20	86.75	0.55***	88.35	86.10	-2.25***
institution attended: technical/vocational (%)	8.40	7.53	-0.86***	6.73	7.86	1.13***
out-of-field degree	4.24	5.30	1.06***	4.57	5.59	1.02***
type of contract: long-term (%)	56.18	64.00	7.82***	68.76	62.13	-6.63***
type of contract: short-term (%)	43.83	36.02	-7.81***	31.26	37.89	6.63***
years of experience	17.83	12.52	-5.31***	13.26	12.22	-1.04***
<i>D. Student Characteristics</i>						
% disadvantaged	73.00	58.25	-14.74***	54.22	59.85	5.64***
mother's years of education	8.96	10.83	1.86***	11.45	10.59	-0.86***
father's years of education	8.86	10.82	1.96***	11.38	10.61	-0.78***
<i>E. Test Scores</i>						
verbal score	-0.43	0.04	0.47***	0.31	-0.07	-0.39***
math score	-0.08	0.34	0.42***	0.59	0.23	-0.36***
Number of Schools	5,113	3,662		1,079	2,583	

Notes: School-level means (2010–2014) by school type: public/municipal, private nonprofit, and private for-profit. Panels A–E cover school characteristics, inputs, teacher characteristics, student background, and SIMCE language/math scores. Monthly tuition is in 2013 CLP.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.2: Effect of For-profit Management on School Inputs

	(1)	(2)	(3)
<i>A. Pupil-teacher ratio</i>			
for-profit	-0.076*** (0.026)	-0.061*** (0.021)	-0.072*** (0.018)
observations	16,802	16,802	11,294
R ²	0.004	0.293	0.327
<i>B. Instruction hours per student</i>			
for-profit	0.111*** (0.032)	0.076*** (0.020)	0.070*** (0.019)
observations	16,802	16,802	11,294
R ²	0.007	0.365	0.414
<i>C. Class size</i>			
for-profit	-0.202*** (0.042)	-0.146*** (0.026)	-0.163*** (0.020)
observations	17,001	17,001	11,342
R ²	0.014	0.430	0.512
municipality $\times$ year FE	no	yes	yes
student controls	no	no	yes

Notes: Results from panel regressions with the log of the column outcome as the dependent variable. Controls: disadvantaged students share, and parental education (mother's and father's years of education). Standard errors clustered at the municipality level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.3: Effect of For-profit Management on School Test Scores

	(1)	(2)	(3)	(4)	(5)
<i>A. Verbal</i>					
for-profit	-0.386*** (0.036)	-0.358*** (0.038)	-0.170*** (0.026)	-0.174*** (0.027)	-0.083*** (0.029)
observations	11,183	11,183	10,204	10,204	10,178
R ²	0.026	0.209	0.437	0.437	0.464
<i>B. Math</i>					
for-profit	-0.355*** (0.038)	-0.336*** (0.038)	-0.148*** (0.027)	-0.150*** (0.027)	-0.036 (0.027)
observations	11,165	11,165	10,194	10,194	10,168
R ²	0.023	0.235	0.473	0.473	0.518
municipality $\times$ year FE	no	yes	yes	yes	yes
student controls	no	no	yes	yes	yes
tuition	no	no	no	yes	no
school inputs	no	no	no	no	yes

Notes: Results from panel regressions; outcomes are standardized test scores (verbal/math) from SIMCE 4th grade, normalized to have mean zero and unit standard deviation in every year. Student controls: disadvantaged students share, mother's and father's years of education. Tuition: annual copayment in 2013 CLP. School inputs: log pupil-teacher ratio, log instruction hours per student, log class size, multigrade classes share, average teacher experience, and share of teachers with an out-of-field diploma. Standard errors clustered at the municipality level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.